

Hello

I'm TS Gughan

I live in chennai and here are my essentials,

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Education

JUL 2024 - 2028

VIT Chennai, Vandalur Kelambakkam Road

B.Tech in Computer Science and Engineering, VIT Chennai (3rd Year)

Skills

- **Programming Languages:** Python, JavaScript, Java, C++, C
- **Web & Full-Stack Development:** React, Django, HTML5, CSS3, Javascript
- **Cloud Infrastructure & DevOps:** Amazon Web Services (AWS Lambda, S3, Cloud Infrastructure)
- **Databases & Data Stores:** SQL, PL/SQL, Supabase, PostgreSQL, Relational Database Management Systems (RDBMS)
- **Specialized Domains:** LLM Finetuning, RAG Applications, AI/ML Integration (Google Gemini API), Web Scraping & Data Pipelines, Object-Oriented Programming (OOP)

Projects (Proof of Work)

- **VoteSmart (v1) | Python, Supabase (PostgreSQL), Web Scraping**
 - Engineered an automated data scraping pipeline in Python to extract public candidate affidavits, assets, and constituency metrics directly from the Election Commission of India (ECI) portal. Designed a normalized relational schema within Supabase to clean, structure, and store highly complex public sector datasets, optimizing indexing for fast real-time user queries. Built a comparative logic engine that processes user-inputted moral and political priorities, cross-references them against the scraped data, and outputs ranked candidate alignment matrices.
- **Donorly | Django, React, PostgreSQL**
 - Developed a social blood donation platform utilizing a React frontend and a Django backend to seamlessly bridge the gap between donors and emergency recipients. Conceptualized and implemented a gamified rewards algorithm ("Hemoglobin Points") to drive recurring user retention and foster local community engagement.
- **Threadly | JavaScript, Real-Time Web**
 - Created an active work forum platform featuring high-speed real-time chatting workflows designed to accelerate team collaboration and task velocity.
- **sumrAi | Python, Google Gemini API**

- Developed an intelligent content simplification app utilizing generative AI to process dense textual documentation and visual images into accessible, context-aware summaries.

- **Web Projects**

- **3D Scan Validator | HTML/JS, CSS, Gemini AI Integration**

- Built an advanced AI-powered image validation tool that evaluates photos intended for 3D body reconstruction. Integrated the Gemini vision API to dynamically analyze user-uploaded images and supply tailored feedback on critical quality criteria such as posture, clothing fit, motion blur, and framing to ensure optimal scan viability.

- **Headphone Recommendation Platform | TypeScript, React, Framer Motion**

- Built a comprehensive audio gear discovery and recommendation web application. Engineered a dynamic, multi-step interactive wizard that parses individual user preferences—including usage scenarios, connectivity, form factor, and budget—to supply tailored headphone matches, complete with side-by-side product comparisons and an administrative management dashboard.

- **Better-Bee | TypeScript, React, AI Integration**

- Built a comprehensive fitness, habit, and calorie tracking web application that provides dynamically updating goal insights. Integrated conversational AI endpoints to parse individual user behaviors and supply tailored workout, meal, and habit adjustments on demand.

- **Wireless Sensor Networks & Optimization Research | MATLAB / NS3, AODV**

Particle Swarm Optimization Architecture Design: Developed a predictive algorithmic framework aimed at maximizing network longevity and Quality of Service (QoS) constraints within bio-medical Wireless Body Area Networks (WBANs).

Algorithmic Optimization: Integrated metaheuristic Particle Swarm Optimization (PSO) with the AODV routing protocol to dynamically select path vectors, successfully mitigating packet congestion and transmission delays.

Analytical Verification: Generated extensive simulation datasets tracking residual energy metrics, packet delivery ratios (PDR), and end-to-end network latency curves.

- **Cloud Interoperability & Portability Research | Kubernetes / Velero, Microservices, Blockchain**

- **Architecture Design:** Developed a modular cloud framework integrating microservices, AI/ML for compliance prediction, and blockchain for immutable security tracking to resolve standardization and interoperability gaps. This design also features a provider-agnostic platform architecture that abstracts the complexities of migrating stateful and stateless workloads across disparate cloud environments.
 - **Algorithmic Optimization:** Orchestrated synchronization agents across source and target clouds using an intermediate S3-compatible transit bucket, dynamically intercepting and remapping cloud-specific storage class requests to standard local equivalents to prevent cluster panic.

- **Analytical Verification:** Extracted real-time telemetry from a baseline migration pipeline (AWS EKS to a local bare-metal cluster) to evaluate data-plane efficiency, generating datasets that tracked a 125-second total execution duration, an 80% resource recreation success rate, and target cluster health statuses.

Experience

June 2026 - PRESENT

DigiTwin, Chennai – *LLM Intern*

- Worked using OCR models, hugging face benchmarking.
- Learned RAG and fine tuning using LoRA and QLoRA of models for specific needs.
- Worked on multi-agent pipelines minimizing compute.
- Worked on pipelines to increase data variance, image resolution and testing.

LEADERSHIP & DOMAIN EXPERIENCE

● **Team Lead** – *AWS Cloud Club, VIT Chennai*

- Direct technical initiatives, hands-on workshops, and collaborative projects for the student community, focusing on AWS serverless compute (Lambda, S3), clean full-stack design patterns, and AI/ML implementations. Leverage design experience in Adobe Photoshop to manage product assets, event documentation, and technical marketing for club-led deployment events.